

Compensation of frequency modulation to amplitude modulation conversion in frequency conversion systems

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Frequency modulation to amplitude modulation (FM-to-AM) conversion is an important issue that can prevent fusion ignition with high power lasers such as the Laser MegaJoule (LMJ). On LMJ, most of the FM-to-AM conversion is expected in the so-called frequency conversion and focusing system, which is a nonlinear system. However, we propose linear transfer functions to compensate the effect of frequency conversion on FM-to-AM conversion. We show that most of AM distortion can be reduced by practical systems: for beam intensity up to 3 GW/cm^2 , the FM-to-AM conversion level can be divided by at least 2, and we almost cancel intensity modulation for intensities below 1 GW/cm^2 . © 2009 Optical Society of America

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1. Introduction

The Laser MegaJoule (LMJ) in France and the National Ignition Facility (NIF) in the United States are high-power laser facilities designed to achieve inertial confinement fusion [1]. Such energetic lasers (up to 2 MJ in a few nanoseconds) have to be phase modulated to broaden the optical spectrum [2,3]. Ideally, this pure phase modulation (i.e., frequency modulation, FM) does not change beam intensity. However, propagation through the different components of the chain is not ideal and slightly filters the optical spectrum. Therefore FM is partly converted into intensity, or amplitude, modulation (AM); this is FM-to-AM conversion [4]. For a fusion experiment and laser-plasma interaction, the control of the temporal shape of the beam pulse is a key point [5]. Intensity distortions and thus FM-to-AM conversion need to be reduced. Recent improvements and studies [6,7] have shown that, on LMJ, we expect most of FM-to-AM conversion in the frequency

conversion and focusing system (SCF). FM-to-AM conversion cancellation is analytically possible for linear systems; we can entirely “compensate” optical filtering by introducing inverse transfer functions. For nonlinear systems such as the SCF, these methods are not intuitive and are not supposed to give as good results as for linear systems.

In the following sections, we present origins of intensity modulation in the SCF and their nonlinear transfer functions. Then we study the performance of possible linear transfer functions to compensate these effects through numerical simulations performed with the Commissariat à l'Énergie Atomique (CEA's) homemade Miró propagation code [8]. Finally, we give leads to a practical implementation for an optimized compensation function for a large intensity panel that could divide at least by 2 the FM-to-AM conversion induced by the SCF on LMJ.

2. FM-to-AM Conversion in Frequency Conversion and Focusing System

On LMJ, the infrared (IR) beam at 1053 nm (1ω) is spectrally broadened thanks to two sinusoidal phase

modulations. If we neglect the slowly varying temporal shape of the pulse, the normalized optical field of the phase modulated beam is given by

$$a_0(t) = \exp\{i[m_1 \sin(2\pi f_{m1}t) + m_2 \sin(2\pi f_{m2}t)]\}. \quad (1)$$

To avoid stimulated Brillouin scattering, $f_{m1} = 2$ GHz and $m_1 = 7$ rad were chosen (anti-Brillouin modulation) [2], and to smooth the beams, $f_{m2} = 14.25$ GHz and $m_2 = 5$ rad were chosen (smoothing modulation) [3]. The associated spectral broadening is given by Carson's rule: 98% of the energy is included in $\Delta f = 2(m+1)f_m$ [9]. When both phase modulations are applied, the spectrum is the convolution of both of them and the spectral width is 203 GHz (32 GHz + 171 GHz), i.e., 0.75 nm at the central wavelength of 1053 nm. In the SCF, the IR beams are converted into ultraviolet (UV) at 351 nm (3ω), and the spectral width is theoretically multiplied by 3 and is thus about 600 GHz.

Phase modulation through propagation is slightly filtered, and these filters modulate the intensity I . To quantify the associated FM-to-AM conversion, the following distortion criterion α is used:

$$\alpha = 2 \frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}}. \quad (2)$$

α ranges from 0 to 200% and is ideally equal to 0 (no intensity modulation). On LMJ, α must be as low as possible (below a few percent). In the following, we only consider the intensity modulation induced by the SCF [7]. The remaining contribution to FM-to-

AM conversion is linear and may therefore be canceled by a linear optical filter.

The SCF, as presented in Fig. 1, is composed of two nonlinear crystals of KDP and DKDP to convert IR beams into UV. Each UV beam has to deliver 7.5 kJ, so the energy conversion efficiency has to be as good as possible (in practice $\sim 80\%$ at maximum intensity). The phase modulated spectrum width is large enough to be partially filtered by the spectral acceptance of the frequency conversion system [see Fig. 2(a)]. To improve the spectral acceptance, a plane grating has been added before the crystals [10–12]. Finally, to focus the beam and to get rid of the unwanted 1ω and 2ω , a focusing grating is used.

As detailed in [7], FM-to-AM conversion has three different origins in the SCF:

- The limited spectral acceptance of the frequency conversion crystals filters the optical spectrum.
- The input intensity modulation is amplified by nonlinearity of frequency conversion.
- The plane grating spatially splits the different wavelengths that follow different path lengths, and so the accumulated phase varies among the pulse spectrum.

For the crystals, as presented in Fig. 2(a), we write the effective baseband transfer function $H_c(f)$ of the 3ω pulse [7] as

$$H_c(f) \approx \text{sinc}(\gamma f) \approx 1 - \frac{\gamma^2}{6} f^2, \quad (3)$$

where γ is linked to the acceptance spectral.

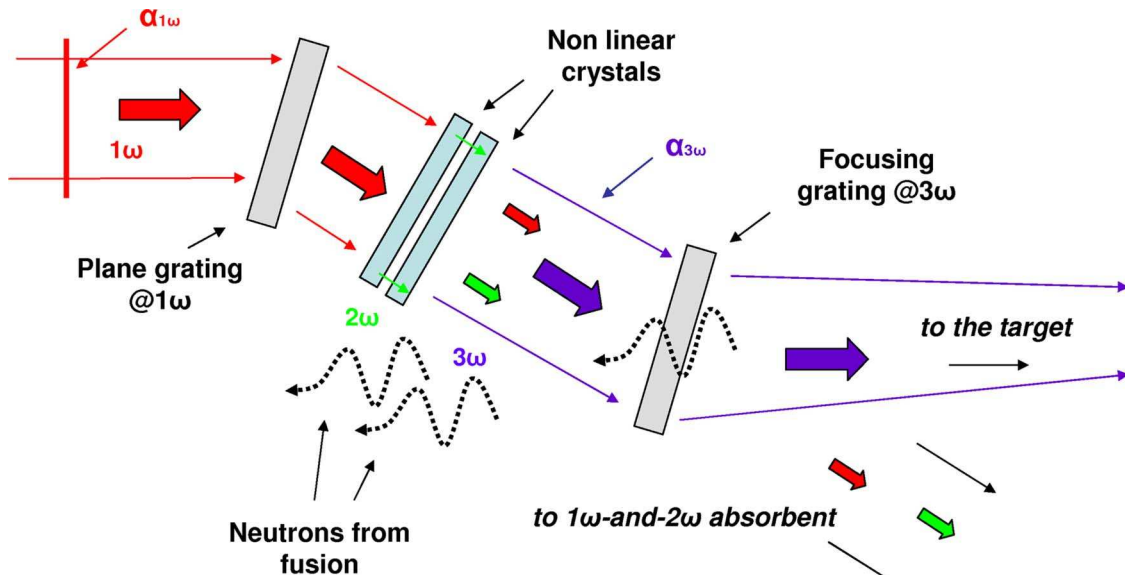


Fig. 1. (Color online) Simple description of frequency conversion and focusing system (SCF) on LMJ. Crystals convert IR into UV thanks to KDP (thickness of 14 mm) and DKDP (10 mm) crystals with a type I–type II configuration. The plane grating (density grooves of 802.7 mm^{-1} with an incident angle of 25°) improves efficiency conversion by dispersion: phase mismatch over the spectrum is considerably lowered. The focusing grating separates UV light from first and second harmonics; it also focuses the beam to the target.

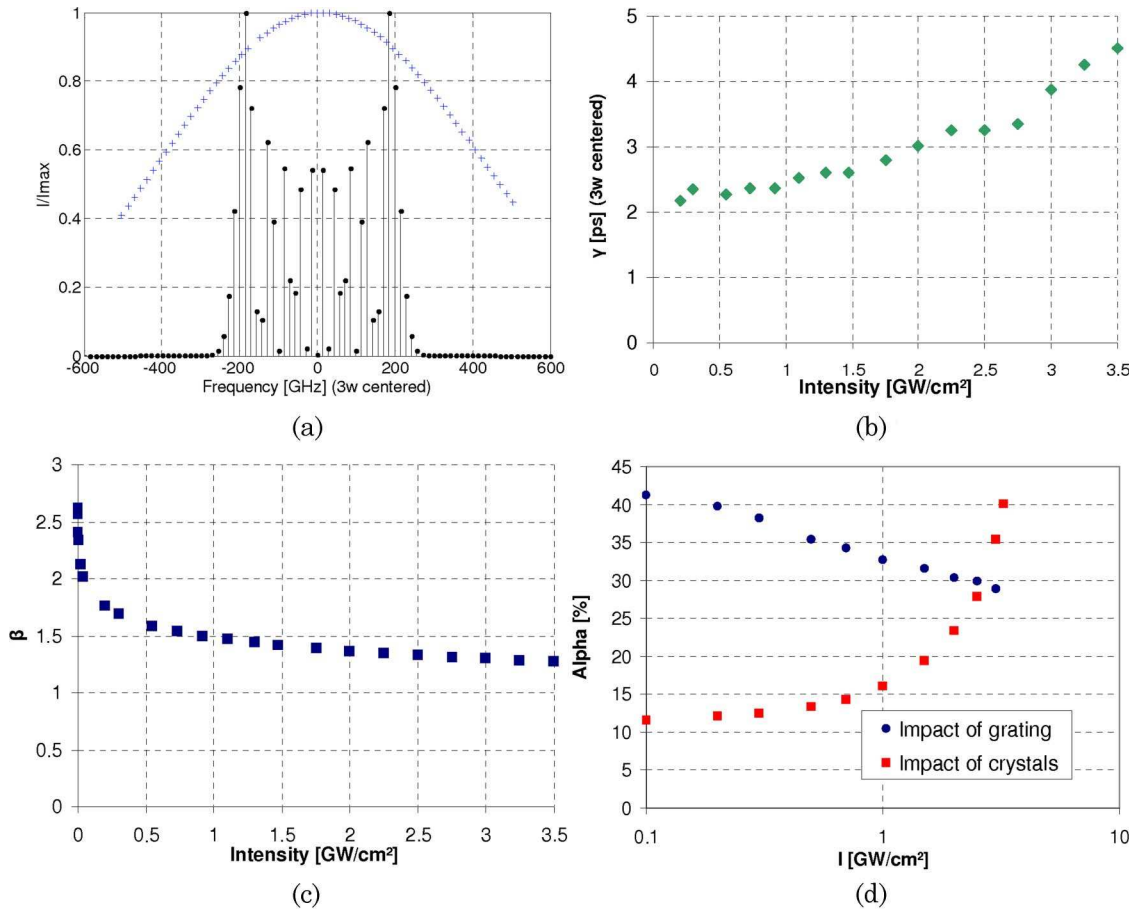


Fig. 2. (Color online) (a) Spectral acceptance at 3ω for the frequency conversion system at 2 GW/cm^2 . It filters the phase-modulated optical spectrum (only the smoothing modulation spectrum is represented, $f_m = 14.25 \text{ GHz}$ with $m = 15 \text{ rad}$). (b) Evolution with intensity of the acceptance spectral parameter γ ; the higher the intensity, the thinner the filter is. (c) Evolution of β defined by $I_{3\omega} \propto I_{1\omega}^\beta$ expressing saturation of frequency conversion. At very low intensity $\beta = 3$. (d) Effect of grating and crystals on FM-to-AM conversion: evolution with the intensity.

This linear model is valid for a constant input intensity. But it is a well known phenomenon that spectral acceptance decreases where intensity increases, and thus γ depends of the initial intensity of the beam as shown in Fig. 2(b) [10], which means (1) the resulting FM-to-AM conversion depends on the considered intensity since the transfer function varies [see Fig. 2(d)], and (2) if there are intensity variations either due to pulse shaping or to previous FM-to-AM conversion, the transfer function varies along the pulse. Note that this approach is theoretically invalid because a linear transfer function in the spectrum domain is supposed to be applicable for an arbitrary long time. Nevertheless, we will show in the following the effectiveness of this approach.

Another nonlinear effect is due to the frequency conversion. Indeed, FM-to-AM conversion at 1ω is amplified by frequency conversion since intensity at 3ω is given by $I_{3\omega} \propto I_{1\omega}^\beta$ with $\beta \leq 3$ because of frequency conversion saturation [10], so a simple first order development of (2) leads to $\alpha_{3\omega} = \beta \alpha_{1\omega}$. β is intensity dependent as presented in Fig. 2(c), and, with Miró, we estimated that β ranges between 1.30 at 3 GW/cm^2 and 1.75 at 0.2 GW/cm^2 .

For the plane gratings, FM-to-AM conversion is due to the accumulated phase φ that is frequency dependent as follows:

$$\varphi(f) = \varphi_0 + \varphi_1 f + \frac{1}{2} \varphi_2 f^2 + o[f^2]. \quad (4)$$

φ_0 and $\varphi_1 f$ do not induce intensity distortions, and so the useful part of the transfer function for FM-to-AM conversion due to dispersion $H_d(f)$ is given by

$$H_d(f) \approx \exp \left[\frac{i}{2} \varphi_2 f^2 \right], \quad (5)$$

where φ_2 depends on the grating, the central wavelength, and the angular configuration [13].

The effect of plane grating on FM-to-AM conversion exists until the focusing grating at 3ω . So Eq. (5) is valid for 1ω and 3ω separately (frequency scale is different for 1ω and 3ω). However, the effect of gratings on FM-to-AM conversion remains intensity dependent because the 1ω effect generates intensity modulation that is amplified by frequency conversion and thus depends on the parameter β that is

intensity dependent as mentioned before [see Fig. 2(d)].

Therefore, even if most of the models to estimate FM-to-AM conversion in the SCF are linear, the global system remains strongly nonlinear through instantaneous variation of γ and β . However, we reduce its effect with linear transfer functions as we show in the following.

3. Reduction of FM-to-AM Conversion by Compensation Transfer Function

The linear transfer functions we are looking for will be precompensation functions because it is easier to act on the optical signal when it is less energetic and where it has a small spatial profile, for example, when the LMJ pulse is still propagated in fibers. So all of the following compensation functions are 1ω centered. We have isolated the causes of intensity modulation from crystals and grating, and thus we first study reduction of FM-to-AM conversion for each of them separately. Finally, we conclude with performances of a global compensation function and practical implementation.

A. Compensation of Plane Grating Dispersion Effects

As discussed in Section 2, the effect of the plane grating occurs both at 1ω and 3ω . Since we aim at precompensating it, the compensation of the 1ω part is straightforward. However, the 3ω part is more complex: precompensation of the pulse distorts the signal, these distortions will be nonlinearly modified by the frequency conversion process, and perfect cancellation is *a priori* not possible. Nevertheless, assuming the shape of a plane grating transfer function given by Eq. (5), we are looking for a function to be applied on the 1ω pulse with a similar expression given by

$$H_{d\text{comp}}(f) = \exp\left[-\frac{i}{2}\varphi_{2\text{comp}}f^2\right], \quad (6)$$

with $\varphi_{2\text{comp}}$ being the function parameter we are looking for.

For LMJ configuration of frequency conversion, total spectral acceptance is defined by both group velocity dispersion (GVD) inside the crystals and plane grating angular dispersion. Indeed, angular dispersion of the optical frequencies of the FM spectrum compensates part of the phase mismatch and thus improves the spectral acceptance [11,12]. With Miró, one can suppress both GVD and angular dependence in crystal without changing anything but the spectral acceptance effect, and so we have performed simulation with a perfect phase match over the whole spectrum. Therefore we perfectly isolate the influence of the grating and we decide to observe the evolution of distortion criterion for different values of $\varphi_{2\text{comp}}$. $\varphi_{2\text{comp}} = 0$ corresponds to no precompensation. Note that α depends on the intensity even for $\varphi_{2\text{comp}} = 0$ as discussed in Section 2.

We can see that, for any considered intensity, it is possible to find a value for $\varphi_{2\text{comp}}$ that almost cancels FM-to-AM conversion as presented in Fig. 3. Optimum value of $\varphi_{2\text{comp}}$ depends on the considered intensity because of the intensity dependence of β , ruling the 1ω amplification of intensity modulation. Effects of gratings can be efficiently compensated thanks to a linear transfer function; we now consider the effect of spectral acceptance and follow a similar approach.

B. Compensation of Spectral Acceptance Transfer Function

We now add GVD and suppress propagation distance between gratings and crystals to suppress the effect of angular dispersion except for its effect on spectral acceptance. Thus we have made numerical simulations with Miró isolating the influence of spectral acceptance, and FM-to-AM conversion is only due to the crystals through their transfer function according to Eq. (3).

Following the idea developed in Subsection 3.A, which has given good results, we are looking for a function to be applied to the 1ω pulse with a similar expression assuming the shape of spectral acceptance transfer function given by Eq. (3) and its approximation:

$$H_{c\text{comp}}(f) = 1 + \frac{\gamma_{\text{comp}}^2}{6}f^2, \quad (7)$$

with γ_{comp} being the function parameter we are looking for.

Parameter γ_{comp} is set as a variable by the software, and we are looking for evolution of the α distortion criterion as presented in Fig. 4. We observe that it is possible to find a value for γ_{comp} that minimizes the distortion criterion induced by spectral acceptance of the frequency conversion system. Note

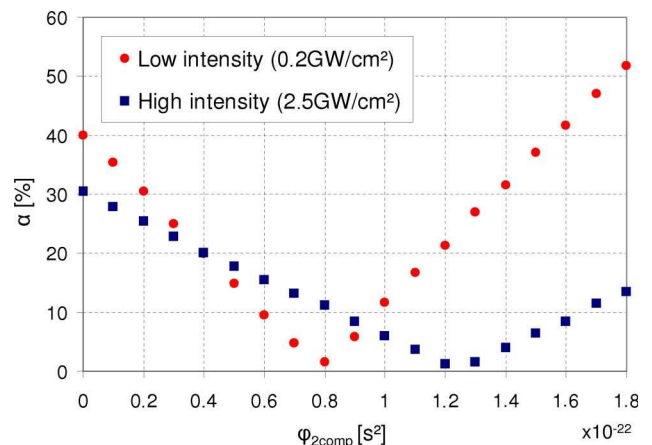


Fig. 3. (Color online) FM-to-AM conversion distortion criterion α for a different compensation function from Eq. (6) considering only the effect of grating angular dispersion (spectral acceptance is set to have a perfect phase matching over the whole spectrum). For any intensity there is an optimized value that almost cancels intensity modulation.

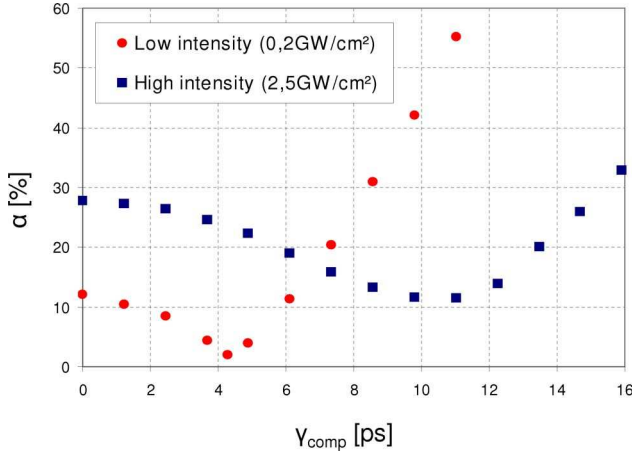


Fig. 4. (Color online) FM-to-AM conversion distortion criterion α for a different compensation function from Eq. (7) considering only the effect of spectral acceptance. For any intensity there is an optimized value that minimizes intensity modulation. The higher the intensity, the less efficient the compensation.

that γ_{comp} cannot be compared to γ because H_c is applied to the 3ω pulse, while $H_{c,\text{comp}}$ is applied to the 1ω pulse. However, when γ increases, γ_{comp} increases too. Instantaneous variation of γ due to intensity modulation prevents perfect compensation and leads to a difference between γ and the optimum value of γ_{comp} for the same intensity. Instantaneous variations also explain that the FM-to-AM conversion reduction is better for low intensity: Figure 2(d) shows that for the crystal transfer function, FM-to-AM conversion distortions strongly vary at high intensity and remain almost constant at low intensity; this instantaneous variation of α cannot be perfectly compensated by a linear transfer function. Nevertheless, the reduction is always significant at any intensity: at 0.2 GW/cm^2 , the distortion criterion is reduced to $\alpha = 2\%$ as compared to $\alpha = 12\%$ without compensation ($\gamma_{\text{comp}} = 0$ in Fig. 3), and at 2.5 GW/cm^2 , the distortion criterion is reduced to $\alpha = 12\%$ as compared to $\alpha = 28\%$ without compensation. Between intensity values of 0.2 and 2.5 GW/cm^2 , reduction of FM-to-AM conversion is better than 50% .

C. Results for Complete Effects in SCF

For both effects of grating and spectral acceptance considered separately, compensation with a linear transfer function leads to an interesting reduction of FM-to-AM conversion distortion. We combine both transfer functions with optimized parameters for any intensity. Because of nonlinearity of the system, it is not obvious that total reduction will be as good as considering both effects separately. In Fig. 5, we have investigated for different intensities the value of distortion criterion for both optimized transfer functions from Eqs. (6) and (7), and we have compared it with the initial distortion criterion. For low intensities, compensation is almost complete: for intensity below 2 GW/cm^2 , α remains under 10% . We notice that for very high intensities, compensation is harder but still significant: the reduction of the distortion

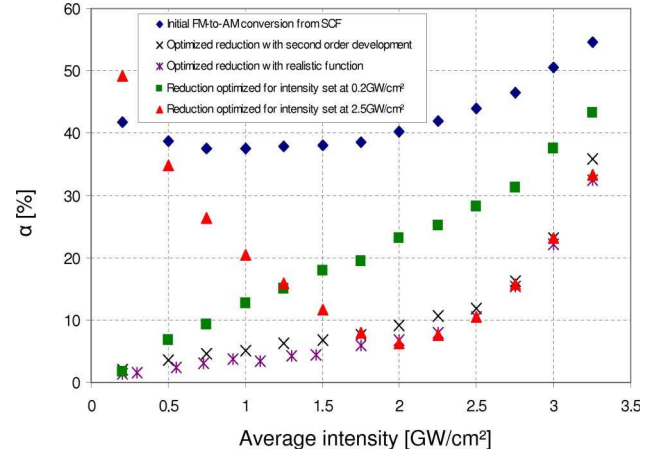


Fig. 5. (Color online) FM-to-AM conversion distortion criterion α for different intensities: (Diamond-shaped points) Intensity modulation for initial system without compensation. (Cross-shaped points) Intensity modulation for optimized compensation systems with a second order transfer function in modulus. (Star-shaped points) Intensity modulation for optimized compensation systems with a realistic system: results are better than a second order system, and reduction is almost complete for intensity below 1.5 GW/cm^2 ($\alpha < 5\%$) and reduces most of FM-to-AM conversion for intensities up to 3 GW/cm^2 . (Square points) FM-to-AM conversion for a SCF with a compensation system optimized at 0.2 GW/cm^2 . At high intensity, intensity modulation are slightly reduced. (Triangle points) FM-to-AM conversion for a SCF with a compensation system optimized at 2.5 GW/cm^2 . At low intensity, intensity modulations are worse than without compensation.

criterion is better than 50% for intensities below 3 GW/cm^2 (close to maximum of intensities considered for LMJ pulses). Simulations predict good compensation performances. However, in practice, $H_{d,\text{comp}}$ and $H_{c,\text{comp}}$ functions are not realistic. In the next subsection, we give leads to a practical implementation of a compensation system in high power lasers.

D. Practical Implementation

To compensate effect of gratings, expression (5) is already a realistic transfer function if we consider chromatic dispersion in fibers [6]:

$$H_{d,\text{comp real}}(f) = \exp \left[-i\pi \frac{\lambda_0^2 D}{c} f^2 \right]. \quad (8)$$

Thus

$$\varphi_{2\text{comp}} = 2\pi \frac{\lambda_0^2 D}{c}. \quad (9)$$

Results from optimization give values for $\varphi_{2\text{comp}}$ between $-0.75 \times 10^{-22} \text{ s}^2$ (for $I = 0.2 \text{ GW/cm}^2$) and $1.3 \times 10^{-22} \text{ s}^2$ (for $I = 2.5 \text{ GW/cm}^2$), so D is between -3.3 ps/nm and -5.5 ps/nm . Hopefully at 1053 nm , chromatic dispersion is negative for standard mono-mode polarization maintaining fibers (PMFs) and we do not need to consider Bragg gratings to compensate dispersion. With experimental measurement from

[6] these values of D are achievable with PMF lengths between 50 and 100 m.

Transfer function (7) cannot be perfectly generated experimentally. Therefore we consider using an interferometric filter to put in the fibered part of the LMJ chain (called the source). Another compensation system using spectral sculpture already exists and could realize the same approximation [14]. The interferometric filter would be generated by a fibered two-wave interferometer composed of an half-wave plate to set the state of polarization at 45° of both axes of a PMF, a polarizer at the output of the fiber, and a second half-wave plate to set the polarization back to the desired state. Then the PMF length sets the delay between the two waves, and the polarizer makes them interfere as in a Mach–Zehnder interferometer. We add a Pelletier cooler to control temperature of the system since constraint stabilization is a key point. This type of filter has been made and is stable [7]. Thus, like any two-wave interferometer, its transfer function is expressed by

$$H_{c\text{real}}(f) = 1 + A \times \exp[i(2\pi f \Delta\tau + \psi)], \quad (10)$$

where $\Delta\tau$ is a time delay proportional to the fiber length, ψ is the recombination phase determined by constraints applied on the fiber, and A is linked to the interferometer contrast that is determined by the angle of the second polarizer.

A second order modulus development of (10) leads to

$$|H_{c\text{real}}(f)|^2 = (1 + A^2) + 2A \cos \psi \left(1 - \frac{(2\pi f \Delta\tau)^2}{2} \right) - 2A \sin \psi (2\pi f \Delta\tau) + o[f^2]. \quad (11)$$

Expression (11) is similar to Eq. (7) if we choose $\psi = 180^\circ$, and so Eq. (11) can be rewritten as

$$|H_{c\text{real}}(f)|^2 = (1 - A)^2 + A(2\pi \Delta\tau)^2 f^2 + o[f^2]. \quad (12)$$

A is set to 0.1 to minimize the energy loss: in that case, the filter energy loss is below 20% [zero order term of Eq.(12)] and can be easily recovered in the source by either an additional fibered amplifier or an amplifier with a slightly higher gain.

With $\psi = 180^\circ$, the expression of the phase from Eq. (10) is an odd function, and thus its development has only odd terms, so there is no additional chromatic dispersion: compensations of both effects from crystals and gratings remain separated. FM-to-AM conversion reduction optimizations have been made with Miró to verify performance of the linear transfer function from Eq. (10). Results presented in Fig. 5 show that it is even better for any intensity, which can be explained by higher order terms in Eq. (10).

The optimization parameter is the PMF length that sets $\Delta\tau$. The lengths are realistic, and if we consider that $\Delta\tau$ is about 1.5 ps/m, the optimized fiber

length is between 0.8 m (for $I = 0.2 \text{ GW/cm}^2$) and 1.8 m (for $I = 2.5 \text{ GW/cm}^2$).

For any considered intensity, it is possible to find a configuration that strongly reduces FM-to-AM conversion due to the SCF. However, in an LMJ pulse, intensity is not constant, so intensity modulation cannot be well compensated in the whole pulse. Figure 5 presents two results considering a compensation optimized for low and high intensities. One can see that a choice has to be made: low intensity compensation does not strongly reduce high intensity AM, and, vice versa, AM at low intensity can even be worse if we compensate high power intensity modulation. For laser plasma interaction, the first part of the ignition pulse at low intensity has to be as stable as possible [15], so FM-to-AM conversion value reduction at low intensity is more interesting: optimization in Fig. 5 shows that intensity modulation can be almost canceled at 0.2 GW/cm^2 ($\alpha < 2\%$). However, for laser damage, reduction of high intensity AM modulation should be more beneficial. A trade-off between both effects has to be made.

4. Conclusion

FM-to-AM conversion in SCF is due to nonlinear phenomena, but it is possible to use linear transfer functions to compensate part of the intensity modulation. For intensities below 3 GW/cm^2 , a realistic configuration can divide by at least 2 the FM-to-AM conversion level. If intensity is constant below 1 GW/cm^2 , distortion criterion is below 5% for an optimized configuration. The drawback of this configuration is that reduction is maximized for a unique intensity, and so a pulse (such as LMJ) with a particular temporal intensity shape is not compensated at any time: choices have to be made considering more disadvantageous intensity modulations.

References

1. www-lmj.cea.fr and www.llnl.gov/nif/project.
2. J. R. Murray, J. Ray Smith, R. B. Ehrlich, D. T. Karazys, C. E. Thompson, T. L. Weiland, and R. B. Wilcox, "Experimental observation and suppression of transverse stimulated Brillouin scattering in large optical components," *J. Opt. Soc. Am. B* **6**, 2402–2411 (1989).
3. J. Garnier, L. Videau, C. Gouédard, and A. Migus, "Statistical analysis for beam smoothing and some applications," *J. Opt. Soc. Am. A* **14**, 1928–1937 (1997).
4. J. E. Rothenberg, D. F. Browning, and R. B. Wilcox, "Issue of FM to AM conversion on the National Ignition Facility," *Proc. SPIE* **3492**, 51–61 (1999).
5. J. D. Lindl, P. Amedt, R. H. Berger, S. G. Glendinning, S. H. Glenzer, S. W. Hann, R. L. Landen, and L. J. Suter, "The physics basis for ignition using indirect-drive targets on National Ignition Facility," *Phys. Plasmas* **11**, 339–491 (2004).
6. D. Penninckx, N. Beck, J.-F. Gleyze, and L. Videau, "Signal propagation over polarization-maintaining fibers: problem and solutions," *J. Lightwave Technol.* **24**, 4197–4207 (2006).
7. S. Hocquet, D. Penninckx, É. Bordenave, C. Gouédard, and Y. Jaouën, "FM-to-AM conversion in high power lasers," *Appl. Opt.* **47**, 3338–3349 (2008).

8. O. Morice, "Miró: Complete modeling and software for pulse amplification and propagation in high-power laser systems," *Opt. Eng.* **42**, 1530–1541 (2003).
9. J. R. Carson, "Notes on the theory of modulation," *Proc. IRE* **10**, 57–64 (1922).
10. A. Boscheron, "Etude de nouvelles configurations de conversion de fréquence pour l'optimisation des lasers de haute puissance," Ph.D. dissertation (Université Paris XI, 1996).
11. S. Skupsky, R. W. Short, T. Kessler, R. S. Craxton, S. Letzring, and J. M. Soures, "Improved laser-beam uniformity using the angular dispersion of frequency-modulated light," *J. Appl. Phys.* **66**, 3456–3462 (1989).
12. V. D. Volosov, S. G. Karpenko, N. E. Kornienko, and V. L. Strishevskii, "Method for compensating the phase-matching dispersion in nonlinear optics," *Sov. J. Quantum Electron.* **4**, 1090–1098 (1975).
13. J. Néauport, N. Blanchot, C. Rouyer, and C. Sauteret, "Chromatism compensation of the PETAL multipetawatt high energy laser," *Appl. Opt.* **46**, 1568–1574 (2007).
14. L. J. Waxer, J. H. Kelly, J. Rothenberg, A. Babushkin, C. Bibeau, A. Bayramian, and S. Payne, "Precision spectral sculpting for narrow-band amplification of broadband frequency-modulated pulses," *Opt. Lett.* **27**, 1427–1429 (2002).
15. "Multiple-FM smoothing by spectral dispersion—an augmented laser speckle smoothing scheme," *Lab. Laser Energetics Rev.* **114**, 73–80 (2008).